INTENSIFIER

JAR INTENSIFIER, HYDRAULIC FLUID SPRING

JARRING

GENERAL DESCRIPTION

HYDRAULIC FLUID SPRING RUN WITH AN OIL JAR TO BOOST JARRING. AS THE STRING IS PULLED, A PISTON COMPRESSES A HIGHLY COMPRESSIBLE SILICONE FLUID AND STORES ENERGY JUST ABOVE THE DRILL COLLARS. WHEN THE JAR TRIPS, THIS ENERGY ACCELERATES THE COLLARS AND JAR UPWARD FOR A HIGH-IMPACT BLOW, LARGELY INDEPENDENT OF STRING DRAG OR DEPTH. IT ALSO CUSHIONS THE REBOUND AFTER EACH STROKE, AND ALLOWS FEWER DRILL COLLARS, ESPECIALLY IN SHALLOW HOLES.

TECHNICAL CHARACTERISTICS

O.D.	I.D.	Total stroke to solid	Collar weight (lb)	Tensile at yield (lb)	Torque at yield (ft-lb)	Fluid volume (gal)
1-5/8	1/4	6	1,100 - 1,400	43,200	260	0.13
1-13/16	5/16	6	1,360 - 1,800	59,400	340	0.195
2-1/4	3/8	6	1,560 - 2,100	118,500	1,800	0.211
2-29/32	1	12-3/4	2,200 - 3,000	194,800	2,260	0.692
3-1/8	1	8-3/4	2,400 - 3,300	229,200	4,068	0.375
3-3/4	1-1/4	8-1/4	4,200 - 5,700	345,000	7,640	0.82
3-3/4	1-1/2	7-7/8	3,400 - 4,600	299,700	5,340	0.63
3-3/4	1-7/8	7-5/8	3,500 - 4,700	179,500	2,980	0.613
4-1/4	1-15/16	8-5/8	3,500 - 4,700	430,300	9,920	0.92
4-1/2	2-3/8	10-3/8	3,600 - 4,900	375,000	11,160	1.15
4-3/4	1-1/2	8-7/8	6,300 - 8,500	591,900	18,420	1.0
4-3/4	2	10-1/8	5,600 - 7,500	468,800	17,200	1.35
6	2	8-5/8	10,200 - 13,800	937,000	34,320	1.57
6-1/4	2-1/4	13	11,800 - 16,000	917,400	40,680	4.24
6-3/4	2-3/8	13	13,000 - 17,500	1,013,800	48,660	3.45
7-3/4	3-1/16	13	11,000 - 15,000	1,587,900	64,020	4.65
9	3-3/4	13	12,000 - 16,000	1,621,000	92,260	3.2

Dimensions in inches. Each intensifier is matched to a jar of the same O.D. x I.D.; connections as for the matching jar. Tensile strengths are calculated theoretical yield values (about +/-20%); torque at yield is theoretical. Figures are a guide only: apply a safety factor.

DESIGN FEATURES

- **Energy storage:** compressible silicone fluid acts as a spring above the collars.
- **Splined mandrel:** torque transmitted in either direction, open or closed.
- **High-pressure seals:** non-extrusion ring assemblies limit seal damage and wear.
- **Safe to fill and use:** fluid-filled, charged with a low-pressure hand pump only.
- **Shock cushioning:** absorbs rebound of the string after each jarring stroke.

OPERATING NOTES

- **String order from the fish:** fishing tool, jar, drill collars, intensifier, string.
- **Stroke:** 6 to 13 in under pull, depending on size.
- **No heavy weights above:** no collars or heavy-weight for at least 1,000 ft above the intensifier.
- **Hot holes:** above 180 F (82 C) bottom-hole temperature, bleed a little fluid before running.

HOW TO ORDER

Please specify: 1. intensifier size and the jar it matches 2. connections and O.D., if not standard 3. spares: packing sets, piston assemblies, seal rings, fill plugs, service kit 4. special intensifier fluid.